



AIR UNIVERSITY

College of Engineering

Department of Electrical Engineering

Course Artificial Intelligence and Neural Networks (lab)
Course code (L): 118920
Instructor: Alina Khan
Pre-requisite(s): ICP
Contact Hours: 3 hours / week

Credit Hours: 2+1

Objectives:

- The aim of this lab is to provide the students with the practical concepts of Artificial Intelligence. Engineers of modern age needs to understand the fundamentals of how an AI system works. From a general purpose computer to an application specific embedded system, AI is involved in all spheres of technology. It is therefore imperative for engineers to have a solid background and hands-experience in development of AI-based algorithms.

Contents:

- Core of the lab course consist of hands-on implementation of various AI algorithms on standard datasets using state-of-the-art Python libraries.

Sustainable Development Goals (SDGs)

Including awareness of specific Sustainable Development Goals (SDGs) is essential in our course, cultivating comprehension and dedication to tackling critical global issues like **SDG 9** and **11**, which focuses on **Industry, Innovation, and Infrastructure** and **Sustainable Cities and Communities**.

Mapping of CLOs and PLOs			
Sr. No	Course Learning Outcomes	PLOs	Bloom's Taxonomy
CLO1	Proceed with AI-tools such as Python for application to AI related problems	PLO5	P2 (Set)
CLO2	Apply standard data preprocessing techniques and local search algorithms to design and develop solutions for complex problems, as well as implement various supervised and unsupervised learning algorithms covered within the course.	PLO3	C3 (Apply)
CLO3	Use effective communication skills both oral and written to defend methodology to solved chosen problem with desired objective	PLO10	A4(Organize Ideas)

CLO Assessment Mechanism					
Assessment tools	CLO1	CLO2	CLO3	Assessment	
Lab Manual	2.25%	25%	0%	30%	
Open Ended Lab	0%	10%	0%	10%	
Lab Exam	0%	30%	0%	30%	
Lab Project / CEA	0%	22%	8%	30%	
Total	2.25%	89.75%	8%	100%	

Grading Policy	
Lab Manual	30%
Open Ended Lab	10%
Lab Exam	30%
Lab Project / CEA	30%

Administrative Instructions:

- According to institute policy, 80% attendance is mandatory to appear in the final examination but 100% will be expected. Approved leaves will not be considered towards attendance.
- Every student should bring calculator and manual in each lab.
- Every student is expected to be in lab before schedule starting time.
- In design-oriented project work, the students deal with problems that can be solved by theories and knowledge they have acquired in their previous lectures. (Design Problems).

Lab Plan

LAB No.	LAB OBJECTIVE	CLOs	Marks	Sign
1.	Introduction to Python: A brief overview of AI computation tools with special Learning focus on python and cloud computing environment	1		
2.	Data Preliminaries: Design and Develop techniques for Preprocessing of raw data including data imputation strategies. Hands-on-activity	2		
3.	Gradient Descent Optimization: Design and Develop Local search using the gradient descent optimization algorithm	2		
4.	The LMS Algorithm: Design and Develop Least mean squares algorithm for iterative learning	2		

5.	Dimensionality Reduction: Principal Component Analysis (PCA) for reducing dimensionality of data	2		
6.	Regression: Design and Develop Model fitting using the least squares regression	2		
7.	The Nearest Neighbor (NN) algorithm: Design and Develop the nearest neighbor and the k-nearest neighbor algorithms	2		
8.	Perceptron Learning Algorithm: Design and Develop Learning linear decision boundaries using the perceptron learning algorithm	2		
9.	Support Vector Machine (SVM): Design and Develop SVM algorithm using the support vectors	2		
10.	K-means Clustering: Design and Develop Unsupervised learning using the K-means clustering algorithm	2		
11.	K-medoids Clustering: Design and Develop Unsupervised learning using the K-medoids clustering algorithm	2		
12.	Feature Selection: Methods for selecting the most appropriate features from data.	2		
13.	Open Ended Lab	2		
14.	Project / CEA Demonstration and Evaluation	2,3		
15.	Semester Lab Exam	2		

Lab	Artificial Intelligence		Student ID	
PLOs	PLO3 – Design and Development		Bloom’s Taxonomy	C3-Apply
	PLO5 – Modern Tool Usage			P2 (Set)
	PLO10 – Communication			A2 (Provide Feedback)
	PLO8 – Ethics			
PERFORMANCE PARAMETERS				
CLOs	Aspects of Assessments	Excellent (75-100%)	Average (50-75%)	Poor (<50%)
CLO1 PLO5	<u>Tools Utilization</u> Apply and discover different basic level functions of Python	Properly and accurately implement the functions and modules of Python as per given task.	Implement the functions and modules of Python with minor errors that give output with slightly differ from the given task.	Does not know how to implement the functions and modules and achieved the required output as describe in a task.
CLO2 PLO3	<u>Design & Development</u> Data Preprocessing Requirement after Problem identification, analysis /literature review, resulting in meaningful conclusions	Completely identifies the Data Preprocessing Requirement for problem in question through efficient analysis/produces near to exact results	Partially identifies the Data Preprocessing Requirement for problem in question and with academic support produces the required results.	Lack of identification of the Data Preprocessing Requirement for problem, needing more than par support to analyze the problem and production of results.
CLO3 PLO3	<u>Design & Development</u> Design / Develop solutions for complex engineering problems covered under the scope of this course	A complete solution / Explain necessary theories according to task description with great use of time and resource material.	Solution was complete but need minor modifications / student could have followed specification more closely.	Solution was complete but did not work, needed several modifications / did not make correct use of resource material or instructions.
CLO4 PLO3	<u>Design & Development</u> Design and Implement different Supervised Learning algorithms using programming concepts and methods and observe the working	A complete solution / Explain necessary theories according to task description with great use of time and resource material.	Solution was complete but need minor modifications / student could have followed specification more closely.	Solution was complete but did not work, needed several modifications / did not make correct use of resource material or instructions.
CLO5 PLO3	<u>Design & Development</u> Design and Implement different Unsupervised Learning Algorithms using programming concepts and methods and observe the	A complete solution / Explain necessary theories according to task description with great use of time and resource material.	Solution was complete but need minor modifications / student could have followed specification more closely.	Solution was complete but did not work, needed several modifications / did not make correct use of resource material or instructions.

	working																	
CLO6 PLO8	Lab Safety Properly handle lab infrastructure/ safety precautions	Properly handle lab equipment & obey safety measures.					Moderate level lab handling and safety measurements					Minor or no safety measurements has been considered.						
CLO7 PLO9	Team Work Completion of Lab tasks with proper team work and contribution.	Proactively work with other team members to complete assigned tasks.					Worked well with team but did not offer much positive feedback.					Very little, if any, contributions to group and less contribution in completion of overall lab tasks.						
EVALUATION																		
CLOs	Aspects of Assessments	1	2	3	4	5	6	7	8	9	10	11	12	13	14	F	P	
CLO1	Recall																	
CLO2	Design and Development																	
CLO3	Design and Development																	
CLO4	Design and Development																	
CLO5	Design and Development																	
CLO6	Lab Safety																	
CLO7	Communication																	
Marks Obtained in Each Lab																		

Signature: _____

Lab Instructor: _____

Signature: _____

Theory Teacher: _____